The Null Cluster as Structural Imprint Recurring Null-Boundary Convergence

A Programme Note after Q-RAIF A

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July 2026

Zenodo DOI: 10.5281/zenodo.20283057

Abstract

Several major structures in relativistic physics are organized at or near null boundaries: causal cones, event and Rindler horizons, horizon thermality, covariant entropy bounds, asymptotic symmetries, soft theorems, memory effects, and the propagation of massless fields. These facts are well known separately. The present note proposes a programme-level unification: these phenomena form a null cluster, a repeated convergence pattern in which causal reachability, thermal response, entropy localization, infrared memory, modular structure, and symmetry constraints co-localize at the null boundary of Lorentzian read-out.

The claim is not that any one of these phenomena is newly derived here. Nor is it that photon or graviton masslessness, BMS symmetry, Unruh thermality, or holographic entropy are new predictions. The contribution is structural: within the Q-RAIF A programme, where a supplied Lorentzian read-out is modelled through algebraic data, the null boundary is read as the thinnest interface of that read-out geometry. It is the place where finite access, causal boundary, state restriction, entropy accounting, and asymptotic imprint most tightly meet.

This note therefore functions as a programme note rather than a proof. It identifies the null cluster, explains why its repeated appearance is not a trivial observation, places it within the Q-RAIF A and mass-energy recovery setting, and states failure modes that would weaken the proposed reading.

Keywords: Q-RAIF A, null boundaries, Lorentzian read-out, horizon thermality, BMS symmetry, soft theorems, memory effect, modular flow, structural imprint

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1 Introduction

This note has a narrow purpose.

It does not introduce a new field equation, a new particle, or a new quantum-gravity model. It does not claim to derive special relativity, the Standard Model mass spectrum, black-hole entropy, soft theorems, or the cosmological constant. It also does not replace ordinary general relativity, quantum field theory, algebraic QFT, or scattering theory.

Its purpose is different:

> to identify a repeated structural convergence around null boundaries and to propose that this convergence is naturally read, within Q-RAIF A, as a structural imprint of Lorentzian read-out.

The central term is **null cluster**.

By the null cluster I mean the recurring co-localization of the following features:

1. causal reachability,
2. horizon or boundary structure,
3. thermal response under restricted access,
4. entropy localization on null or near-null sheets,
5. infrared memory and asymptotic symmetry,
6. modular flow with geometric meaning in special cases,
7. massless or near-massless propagation as a kinematic locking to the null cone.

Not every feature appears in every physical situation. The claim is weaker and more useful:

> whenever a physical regime is organized by a null boundary, several of these features tend to reappear together.

This is not a theorem. It is a programme-level organizing principle. This note commits to the weak form: the cluster is a reproducible taxonomy worth testing. It does not claim that one Q-RAIF mechanism generates all nodes. The strong common-origin form would require the minimal-sufficiency model defined below and remains open.

2 What Is Already Known

The individual ingredients of the null cluster are standard.

The causal cone of Lorentzian geometry is defined by the null directions. Event horizons are null hypersurfaces. Rindler horizons produce observer-dependent thermality. Black-hole entropy scales with horizon area. Bousso's covariant entropy bound uses null light-sheets. Asymptotically flat gravity has null infinity, BMS symmetries, soft graviton theorems, and gravitational memory. In algebraic QFT, the Bisognano-Wichmann theorem connects modular flow for wedge algebras with Lorentz boosts. In high-energy or massless limits, propagation is locked to the null cone.

None of that is new.

The useful question is different:

> Why do these structures repeatedly meet at null boundaries?

Mainstream literature usually studies partial clusters. The references in the table are representative anchors, not exhaustive bibliographies.

Line of work	**Representative anchor**	**Null phenomena emphasized**	**Usually outside its main focus**
BMS / infrared triangle	Strominger 2018	Asymptotic symmetry, soft theorems, memory	Horizon thermality, entropy bounds, modular flow
Thermodynamic gravity	Jacobson 1995	Local horizons, heat, Einstein equation	BMS, soft theorems, memory
Covariant entropy bounds	Bousso 1999	Null light-sheets, entropy limits	Modular theory, BMS, massless protection
Algebraic QFT / modular theory	Bisognano–Wichmann 1975; Haag 1996	Wedge modular flow, Rindler thermality	BMS, covariant entropy bounds
Conformal / twistor approaches	Penrose 2010	Null geometry, conformal structure	Horizon thermality, entropy localization
Gauge and diffeomorphism protection	Weinberg 1965; Wald 1984	Massless propagation, constraint structure	Horizon thermality, entropy bounds, soft memory

The present note proposes that these partial clusters are not merely parallel accidents. They are different faces of a broader null cluster.

3 Why the Observation Is Not Trivial

It is tempting to say: "Of course these things involve light cones. This is just relativity."

That response misses the point.

The nontrivial point is not that light cones exist. The nontrivial point is that multiple independent-looking structures concentrate at the same kind of boundary:

- causal structure says what can influence what;
- horizon thermality says restricted access to a wedge or exterior region produces a thermal state;
- entropy bounds say information is controlled by area or null sheets, not by naive volume;
- soft theorems and memory say radiation leaves persistent boundary data at null infinity;
- modular theory says, in important cases, state-dependent algebraic flow can acquire geometric boost meaning;
- massless propagation says protected gauge or diffeomorphism structures lock certain excitations to the null cone.

These are not the same statement in different notation.

They are also not all derivable from the causal cone alone. A null cone plus a Lorentzian metric does not select a quantum state, a wedge restriction, a temperature, an entropy functional, an asymptotic phase space, or a modular operator. Horizon thermality additionally uses a state and restricted observer algebra; entropy bounds use gravitational dynamics and entropy assumptions; BMS memory uses asymptotic field equations and scattering data; modular geometry uses a von Neumann algebra and cyclic/separating state. These extra inputs show that the nodes contain more structure than the single fact $ds^2 = 0$.

The converse caution is equally important: their common localization may still follow from ordinary Lorentzian causal geometry plus those separate inputs. This note does not yet prove an additional Q-RAIF mechanism linking them. Its weak claim is taxonomic; the stronger common-origin claim remains open.

They come from different parts of physics:

- Lorentzian geometry,
- general relativity,
- quantum field theory,
- algebraic QFT,
- black-hole thermodynamics,
- scattering theory,
- asymptotic symmetry,
- gauge theory.

The priority value of this note is to draw these lines on one map and ask whether their common null locus is a structural clue.

4 The Null Cluster

4.1 Core Cluster

The core cluster consists of phenomena that directly occur on, or are defined by, null boundaries.

Node	Standard meaning	Null locus	Programme reading
Causal cone	Boundary between timelike and spacelike separation	$ds^2 = 0$	Basic boundary of effective read-out
Event horizon	Boundary of causal access	Null hypersurface	Finite access boundary in geometry
Rindler horizon	Acceleration horizon	Null planes	Restricted wedge access yields thermality
Horizon thermality	Unruh / Hawking type thermal response	Horizon-local	State restriction at null boundary
Entropy bounds	Area or light-sheet control of entropy	Null sheets / horizons	Finite read-out closes on boundary data
BMS / soft / memory	Asymptotic symmetry and persistent radiation imprint	Null infinity	Null boundary stores scattering imprint
Modular geometry	Modular flow becomes boost in special QFT settings	Wedge / horizon structure	Algebraic flow can carry geometric boundary meaning

These are the strongest candidates for the null cluster proper.

4.2 Adjacent Node: Massless Protection

Massless protection should not be treated as identical to the core cluster.

Photon masslessness is protected by gauge structure. Graviton masslessness, in the usual low-energy picture, is tied to diffeomorphism structure and massless spin-2 consistency. These are not the same mechanisms as horizon thermality, entropy bounds, or BMS memory.

Still, they belong near the cluster.

The reason is **kinematic locking**:

> gauge and diffeomorphism structures protect the absence of mass terms, and the absence of mass locks propagation to the Lorentzian null cone.

Thus massless protection is not a core null-boundary phenomenon in the same sense as a horizon or null infinity. It is an adjacent node: a symmetry-protected route by which field propagation becomes null.

This distinction matters. If massless protection is placed inside the core cluster too quickly, the note overclaims. If it is excluded completely, the relation between symmetry protection and null propagation is lost.

The correct placement is second ring, not first ring.

4.3 Inclusion and Exclusion Criteria

The cluster is not meant to include every use of null geometry.

A proposed blind test for the **core cluster** has three parts:

1. a null hypersurface or null asymptotic locus is part of the input data;
2. the observable, state restriction, flux, or entropy quantity is defined on that locus; and
3. replacing it by a generic timelike or spacelike locus while holding the other inputs fixed changes or invalidates the construction.

This test could in principle admit examples not chosen in advance.

A phenomenon belongs to the **adjacent ring** if it fails the boundary-localization test but a declared symmetry or field equation enforces the on-shell condition $q(p) = 0$. Massless protection is the main example: gauge and diffeomorphism structures do not equal horizon thermality or BMS memory, but they can constrain propagation to the null cone.

A phenomenon is **outside the cluster** if it fails both tests. Twistor, conformal, causal-set, or celestial constructions are not excluded by name; they must be classified by the same operational test. The taxonomy remains heuristic until "changes or invalidates" is made quantitative in concrete models.

This test is meant to reduce cherry-picking. The cluster is not "everything involving light cones"; it is the narrower convergence of boundary, access, entropy, memory, modular, and symmetry-locking structures around null loci.

5 The Structural-Imprint Claim

The programme-level claim can now be stated.

> The null cluster is a structural imprint of Lorentzian read-out: it marks the boundary where finite causal access, state restriction, entropy accounting, asymptotic radiation data, and symmetry-protected massless propagation most tightly meet.

Revised Q-RAIF A treats a Clifford representation as a model for a supplied Lorentzian read-out; it does not derive that read-out from the algebraic constraints. The mass-energy companion then checks the standard mass-shell identity after Lorentzian and translation/Stone bridge data are assumed.

The present note takes a further programme-level step:

if Lorentzian read-out is the arena, then null boundaries are candidate interfaces on which several kinds of restricted data can be compared.

This is why the same class of features keeps recurring:

- causal closure,
- horizon restriction,
- thermal response,
- entropy localization,
- soft boundary data,
- memory,
- modular-geometric flow,
- massless propagation.

The phrases "thinnest interface" and "boundary grammar" require an operational meaning. For a target family of read-out observables $\mathfrak{O}$ and tolerance ε, call boundary data $D(N)$ on a null locus N *minimal sufficient* only if (i) $D(N)$ reconstructs $\mathfrak{O}$ to accuracy ε, (ii) no proper declared sub-datum does so, and (iii) comparable data on generic timelike or spacelike loci require strictly more resources under a specified cost. This criterion could have selected another causal type.

No model in this note proves conditions (i)–(iii). Consequently the null locus is a candidate minimal interface, not an established "boundary grammar of Lorentzian manifestation." Constructing and comparing the required reconstruction costs is the true technical task behind the metaphor.

6 Pairwise Structural Links

The null cluster becomes more visible when the pairwise links are made explicit.

6.1 Causal Boundary and Horizon

A horizon is not a material wall. It is a causal boundary. Its definition depends on which signals can reach which region, and this is organized by null curves or null hypersurfaces.

Thus the causal cone and horizon structure are not separate topics. A horizon is what the causal cone becomes when access is globally or regionally restricted.

6.2 Horizon and Thermality

The Unruh and Hawking effects show that horizon-related access restrictions are tied to thermal response.

In a Rindler wedge, the restriction of the vacuum to the wedge is thermal with respect to boost time. In black-hole settings, the horizon is tied to Hawking temperature. The details differ, but the repeated pattern is the same: restricted access at a horizon gives a thermal reading.

Programme reading:

> thermal response is not added on top of a horizon; it is one of the signatures of finite access to a null-bounded region.

6.3 Horizon and Entropy

Black-hole entropy and covariant entropy bounds indicate that entropy is controlled by boundary geometry, especially null or horizon-like structure.

The important shift is from volume to boundary.

Programme reading:

> when finite read-out closes a region, the entropy-relevant data are organized at the boundary of access, not throughout a naive container volume.

The quantum focusing conjecture and related quantum Bousso-bound work sharpen this line by replacing purely classical area focusing with generalized entropy and quantum expansion along null congruences. This does not prove the present note's imprint reading, but it reinforces the point that entropy constraints and null deformation data are not separate bookkeeping devices.

6.4 Null Infinity and Memory

In scattering theory, radiation reaches null infinity. Soft theorems, BMS symmetries, and memory effects show that null infinity is not empty bookkeeping. It carries persistent imprint.

Programme reading:

> null infinity is where scattering leaves boundary-readable memory.

This is the asymptotic analogue of the horizon claim. Both cases treat null boundaries as places where finite read-out becomes constrained by persistent boundary data.

Recent work on BMS charge algebras, double-soft theorems, BMS fluxes, and celestial holography makes this point more concrete: null infinity is not merely a formal endpoint of radiation, but a boundary arena where flux, symmetry, memory, and candidate celestial operator data are organized.

Celestial holography is also a competing explanation, not merely supporting evidence: it offers a calculational programme in which bulk scattering data are reorganized as boundary correlator data at null infinity without invoking Q-RAIF A. The present note adds only a cross-domain taxonomy and the proposed minimal-interface test. Until that test yields a different prediction or compression bound, the imprint language has no demonstrated empirical or mathematical advantage over the celestial account.

6.5 Modular Flow and Null Boundaries

The Bisognano-Wichmann theorem shows that, in relativistic QFT under suitable assumptions, the modular flow of a wedge algebra is implemented by Lorentz boosts.

This does not mean modular flow is always physical time, nor that modular flow directly defines momentum.

The narrower lesson is enough:

> in important cases, state-dependent algebraic flow knows about geometric wedge and horizon structure.

This is precisely the kind of link Q-RAIF A treats as structurally significant, while still avoiding the overclaim that modular flow alone derives all spacetime dynamics.

The limitation is important. The Bisognano-Wichmann result is a statement about wedge algebras in relativistic QFT under specific structural assumptions; it is not a generic theorem about arbitrary null surfaces, arbitrary states, or curved spacetime. Interacting theories, Haag duality issues, Type III factors, and the detailed role of Connes spectrum belong to a deeper AQFT analysis that this note does not attempt.

There is, however, targeted recent work closer to the present boundary language: studies of modular operators for null-plane algebras in free fields decompose the one-particle structure into lightlike fibres and relate null cuts, relative entropy, and QNEC-type statements. This supports the limited claim made here: modular data can have genuine null-boundary content, but the technical domain of validity must be stated explicitly.

6.6 Massless Propagation and Null Cone

A massless excitation propagates on the null cone. In gauge theory, the absence of a photon mass is protected by gauge invariance. In gravity, the masslessness of the low-energy spin-2 mode is tied to diffeomorphism structure and consistency constraints.

Programme reading:

> massless propagation is the field-theoretic way a symmetry-protected degree of freedom is locked to the Lorentzian boundary.

This is adjacent to, but not identical with, the horizon and entropy parts of the cluster.

7 Placement after Q-RAIF A

This note assumes the Lorentzian arena modelled by revised Q-RAIF A:

> an effective sector is supplied with Lorentzian geometry and may use a Clifford representation to encode its tangent form.

The mass-energy companion paper adds:

> once Lorentzian and admissible four-momentum bridge data are supplied, the standard mass-energy relation follows as a definitional identity.

This note adds a programme-level map:

> once Lorentzian read-out is available, the null boundary becomes the natural interface where causal, thermal, entropic, asymptotic, modular, and massless-propagation features repeatedly concentrate.

The three statements have different epistemic status.

Claim	**Status**
Q-RAIF A Lorentzian read-out	Interface/model input
Mass-energy recovery	Definitional identity given Lorentzian and momentum bridges
Null cluster as structural imprint	Weak taxonomic programme claim

This separation is essential. The present note should not be cited as proving the null cluster or as inheriting a Lorentzian-emergence theorem. It proposes a weak taxonomy and a research test.

8 What Is New

The note does not claim novelty for the following:

- light cones,
- event horizons,
- Hawking or Unruh thermality,
- black-hole entropy,
- covariant entropy bounds,
- BMS symmetry,
- soft theorems,
- memory effects,
- massless photon or graviton propagation,

- modular theory in AQFT.

The novelty is the cross-domain placement.
The note says:

1. these are not merely isolated null-related facts;
2. they form a repeated cluster around null boundaries;
3. the proposed minimal-sufficiency test asks whether null boundaries compress the relevant read-out data better than timelike or spacelike alternatives;
4. the cluster suggests specific failure modes and future directions.

This is a weak perspective contribution, not a new calculation or a common-origin result.

9 Failure Modes

The following would weaken or falsify the proposed reading.

The statuses are not equal. F1–F4 primarily falsify inherited ingredients of a single Lorentzian-cone or standard mass-shell description; they would matter even if the null-cluster taxonomy had never been proposed. Only F5 directly tests this note's added claim that the listed nodes form a useful relation beyond their separate standard explanations.

9.1 F1. Fundamental vacuum dispersion

If future observations establish a fundamental, non-environmental, energy-dependent speed of massless propagation in vacuum, then the idea of a single effective Lorentzian null cone would be weakened.

This would not automatically refute every algebraic-emergence programme, but it would force Q-RAIF A to be modified toward multi-cone, Finsler-like, Lorentz-violating, or energy-dependent read-out geometry.

The relevant target is not ordinary plasma dispersion, gravitational Shapiro delay, source-intrinsic time lag, or effective propagation in matter. The dangerous signal would be a vacuum law of the form

$$v(E) = c\left[1 + O\left(\left(\frac{E}{E_{\mathrm{QG}}}\right)^{n}\right)\right]$$

or another non-environmental Lorentz-violating dispersion that survives multi-messenger and multi-frequency controls. In such a case the note's single-null-cone reading would have to be weakened or replaced.

9.2 F2. Basic photon mass

If the photon is found to have a fundamental nonzero rest mass, not merely an effective mass in a medium or plasma, then the role of the photon as the cleanest electromagnetic null pointer would fail.

The broader null cluster could survive, but its most familiar massless representative would be downgraded.

9.3 F3. Basic graviton mass

If gravitational waves are shown to arise from a fundamentally massive spin-2 mode, rather than a massless geometric perturbation or an effective modification, then the placement of gravitational waves as direct null-boundary geometry would fail.

9.4 F4. Mass-shell failure inside a valid Lorentzian sector

If a system has:

- a valid effective Lorentzian geometry,
- a well-defined four-momentum,
- and ordinary energy-momentum read-out,

but violates the standard mass-shell relation in a way that cannot be attributed to medium effects, interactions, curved-background corrections, or bad variable choice, then the mass-energy companion bridge would be undermined.

9.5 F5. No deep relation among horizon thermality, entropy, memory, and modular structure

If future work showed that horizon thermality, entropy localization, infrared memory, and modular geometry are only unrelated special-case coincidences, then the null cluster would lose its unifying force.

This is the direct failure mode for the note. It would not be a single experimental refutation, but a failure of the taxonomy and minimal-interface programme to generate relations or predictions beyond ordinary Lorentzian causal structure plus separate physical inputs.

10 Open Directions

10.1 Null Boundary as Imprint Interface

The most publishable next technical direction is to connect BMS symmetry, soft theorems, memory effects, and imprint language.

The guiding idea:

> null infinity is not only where radiation escapes; it is where scattering leaves boundary-readable imprint.

This can potentially be written as a technical essay with standard references and without extending Q-RAIF machinery.

10.2 Cosmological Constant as Global Read-out Curvature

A second direction is the cosmological constant.

The cautious claim:

> the cosmological constant problem may be partly a category problem if one treats global read-out curvature as ordinary local vacuum energy density.

This is not a replacement for Lambda-CDM. It is not a solution to the numerical problem. It is a programme-level reframing.

10.3 Black-Hole Information as Read-out Sector Reallocation

The black-hole information problem can be cautiously read as a mismatch between sector-limited external read-out and global algebraic accounting.

Modern island and replica-wormhole results already suggest that the accessible description of entropy depends on how the gravitational and radiation sectors are assigned.

The programme-level reading:

> the question is not simply where information goes, but which read-out sector can access which imprint.

10.4 Neutrinos near the Null Boundary

Neutrino masses are extremely small compared with other known fermion masses. This makes them natural candidates for a "near-null" structural reading.

This should remain a weak direction. Seesaw mechanisms, Majorana versus Dirac mass, sterile neutrino constraints, and mass ordering are concrete physics questions. The programme should not replace them with metaphor.

The only safe statement is:

> neutrinos may be an example where a tiny timelike thickness sits very close to the null class.

10.5 Dark Sector as Limited Observable Access

The dark sector is the riskiest direction.

It is tempting to say that dark matter is not a particle but a read-out effect. That is too strong.

The publishable version is narrower:

> some dark-sector phenomena may be usefully interpreted as effects of limited observable access, where gravitational read-out is broader than electromagnetic read-out.

This remains an open conjecture. It requires quantitative modelling before it can become a prediction.

11 Limits

This note has strict limits.

1. It does not derive the null cluster.
2. It does not derive BMS symmetry, soft theorems, memory effects, horizon entropy, or modular flow.
3. It does not claim that all null-related phenomena have the same physical mechanism.
4. It does not treat gauge invariance and diffeomorphism invariance as the same mechanism.
5. It does not claim that dark matter is not a particle.
6. It does not solve the cosmological constant problem.
7. It does not solve black-hole information.
8. It does not introduce religious or metaphysical terminology into the physics note.

The correct label is:

> structural programme note.

12 Minimal Claim

The minimal claim is this:

Null boundaries repeatedly gather causal limits, horizon restriction, thermal response, entropy localization, asymptotic memory, modular geometry, and symmetry-protected massless propagation.

Q-RAIF A gives this recurrence a programme-level placement: the null boundary is the thinnest interface of Lorentzian read-out, where finite access and boundary data meet.

This is not a new equation; it is a map.

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